Market Power and Contagion in Interbank Networks

An Agent-Based Model of Systemic Risk

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Outline

1. Abstract
2. Risk and Connectivity
3. Model Framework
4. Interest Rate and Market Power
5. Equity and Bankruptcy Dynamics
6. Phase Transition
7. Results
8. Conclusions

Abstract: ABM, Interbank Market, and Contagion

- We use **Agent-Based Models** to study systemic risk in interbank credit networks.
- The dynamics are driven by two key parameters: the **probability of attachment** p Erdős–Rényi graphs as interbank connections generator (exogenous) and the **interest rate** r (endogenous).
- As p increases, the network becomes more connected, forming a giant component. Interest rates decline due to greater competition and reduced market power.
- However, higher connectivity also raises the probability of borrower bankruptcy. Excessively low interest rates, maintained for too long, can be **detrimental to the stability of the system.**

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The 2008 Crisis: Background

- A prolonged period of low interest rates encouraged excessive risk-taking and increased bank leverage.
- Standard competitive equilibrium models failed to capture the complex interactions that amplify crises.
- The interbank market became a **contagion vector**: failure of one bank triggered a domino effect through interlocking exposures, revealing how interconnectedness can magnify systemic fragility.

The Risk-Sharing Trade-off

Two opposing effects of network connectivity:

- **Benefit:** Increasing connectivity allows **risk sharing**—banks diversify exposure across more counterparties, lowering interest rates.
- **Cost:** Excessive connectivity increases **systemic risk**—default propagates through the network, and lenders serving a larger number of clients face higher bankruptcy exposure.
- Evidence: a **non-monotonic relationship** exists—low connectivity stabilizes by diluting shocks, but high connectivity facilitates cascading failures. Taken together, these effects imply that excessively low interest rates increase both bankruptcies and contagion.

Failure Channels in Financial Crises

In literature, three primary propagation mechanisms of systemic failure are identified:

1. **Bank runs:** self-fulfilling panics where depositors withdraw mass funds simultaneously.
2. **Asset price contagion:** fire-sale spirals that depreciate assets across the system.
3. **Interlocking exposures:** direct linkages through the interbank market—default of one bank triggers defaults in connected banks.

Failure Channels in the Model

In this model, we consider the three failure channels:

- **Rationing** (insufficient funding of borrower),
- **Contagion** (lender suffers losses from borrower default),
- **Repayment failure** (borrower cannot repay loan).

At low connectivity: rationing dominates; at high connectivity, repayment failure becomes the most relevant.

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Model Overview

- ABM with N banks in a sequential economy $t = \{0, 1, \dots, T\}$.
- Banks connected through an **Erdős–Rényi** random graph $G(N, p)$: each edge exists with exogenous probability p (probability of attachment).
- Goal: analyze how **connectivity** (p) and **market power** (ψ) affect system stability through the endogenous adjustment of r.

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Balance Sheet

Accounting Identity

$$L_i^t + C_i^t + R_i^t = D_i^t + E_i^t$$

- **Assets:** Long-term assets (L), liquidity (C), reserves (R).
- **Liabilities:** Deposits (D), equity (E).
- Reserves set by regulation: $R = \hat{r}D$ with $\hat{r} = 0.02$.

Liquidity Shocks

Deposits are exogenously shocked each period:

$$D_i^t = D_i^{t-1}[\mu + \omega \cdot U(0,1)]$$

- $\mu = 0.7$: expected growth rate.
- $\omega = 0.35$: shock magnitude.
- **Potential lenders:** banks with supply of C (after rebalance).
- **Potential borrowers:** banks with demand of loan ($C < 0$).

Network Structure

- Bilateral credit lines formed at the start of each period.
- Connectivity controlled by parameter p in the Erdős–Rényi graph.
- Each borrower is matched to **exactly one** lender per period.
- The **Giant Component Size (GCS)** measures network integration.

Erdős–Rényi Random Graph

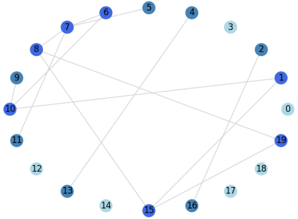
$p = 0$ $p = 0.1$ $p = 0.2$

Each of the $\binom{N}{2}$ edges is included independently with prob. p. For $N = 20$ banks:

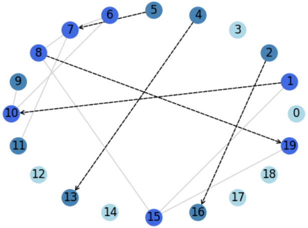
- $p = 0$: isolated banks, no interbank market.
- $p = 1/N = 0.05$: **phase transition**: giant component begins to emerge.
- $p > 0.105$: network is saturated ($> 92\%$ of banks connected): **supercritical regime**.

Why supercritical is relevant in this model

Erdős–Rényi generated for $N = 20$



Considered links after applying restrictions



Network Structure

The Erdős–Rényi model does not attempt to replicate the entire interbank market; rather, it illustrates how connection density alters the probability and scale of contagion.

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Expected Profit

Banks are risk-neutral in perfect competition. Expected profit for lender i from lending to borrower j:

$$\mathbb{E}[\Pi_t^{i,j}] = \underbrace{p_t^j(r_t^{i,j} c_t^{i,j} - \overbrace{m\, c_t^{i,j}}^{\text{cost of loan}})}_{\text{normally pays}} - \underbrace{(1 - p_t^j)(m\, c_t^{i,j})}_{\text{when fails}}$$

- m: screening cost (0.015).
- p_j^t: probability of borrower survival.
- $r_{i,j}^t$: interest rate; $c_{i,j}^t$: lending capacity.

Interest Rate Derivation

Setting $\mathbb{E}[\Pi] = 0$ (zero-profit condition):

$$r_{i,j}^t = \frac{m}{p_j^t}$$

Introducing **market power** $\psi \in [0, 1]$:

$$r_{i,j}^t = \frac{m}{p_j^t(1-\psi)}$$

- $\psi = 0$: competitive rate.
- As connectivity p increases, competition intensifies, market power ψ declines, and the interest rate falls.

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Endogenous Market Power

Market power evolves endogenously with bank size:

$$\psi_i^t = \frac{E_i^t}{E_i^{t,\max}}$$

- Larger banks (higher equity) exercise more market power.
- Borrower default probability: $p_{b,j}^t = E_j^t / E_j^{t,\max}$.
- As p increases, the market becomes more competitive — borrowers can choose among more lenders, reducing lenders' market power. However, this also allows lenders to serve more clients, increasing their exposure to borrower default.

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Equity Evolution

$$E_i^t = E_i^{t-1} + \sum_j l_{i,j}^t r_{i,j}^t - \sum_{j \subseteq \theta_i^t} \left[B_{i,j}^t - \tilde{L}_j^t \right]$$

- **Bankruptcy:** bank exits if $E_i^t < 0$.
- **Fire sales:** last-resort option (blocked in this model to amplify contagion effects).
- Failed banks replaced by new entrants with small initial capital.
- Endogenous interest rates affect bank earnings; low rates compress margins but also reduce the cost of borrowing, generating opposing effects on stability.

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Failure Channels

Three distinct causes of bank failure:

1. **Rationing:** borrower cannot secure sufficient funding.
2. **Contagion:** lender suffers losses from borrower default (bad debt).
3. **Repayment failure:** fully-funded borrower cannot repay loan or interest.

Rationing dominates at low connectivity (fewer lending opportunities). **Repayment failure dominates** at high connectivity, as lenders serve more clients and the probability of borrower bankruptcy rises with lower interest rates.

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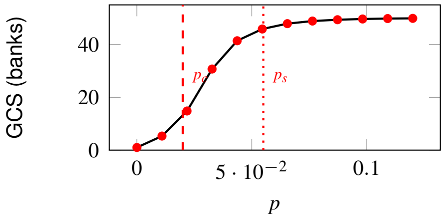
Phase Transition

- **Phase transition:** GCS jumps from ~ 1 to ~ 46 banks (92%) as p crosses the supercritical region (above $p_c = 1/N$).
- **Low p:** rationing-dominated, high interest rates ($r \approx 3.0$), few loans granted.
- **High p:** repayment-failure-dominated, low interest rates ($r \approx 1.8$), saturated network.
- As the market becomes more competitive, lenders serve more clients, the probability of bankruptcy rises, and bad debt accumulates.

Phase Transition



$N = 50$ banks, $\langle k \rangle = p(N-1)$

Percolation threshold $\langle k \rangle \approx 1$ (phase transition) for Erdős–Rényi graph:

$$p_c = \frac{1}{N} = \frac{1}{50} = 0.02$$

Supercritical region $\langle k \rangle = 2.14$, observed at $p_s = 0.055$.

p	GCS
0.00001	1
0.011	5
0.022	**15**
0.033	31
0.044	41
0.055	**46**
0.065	48
0.087	49
≥ 0.11	50

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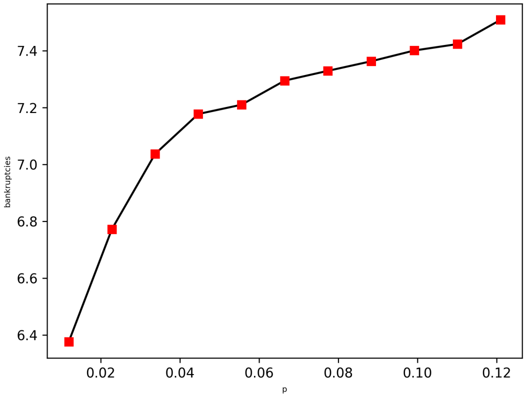
Bankruptcies



Average number of failures during simulations.

p	bankruptcies
0.012	6.3762
0.023	6.7724
0.034	7.0369
0.045	7.1774
0.056	7.2100
0.066	7.2948
0.077	7.3292
0.088	7.3628
0.099	7.4007
0.110	7.4234
0.121	7.5088

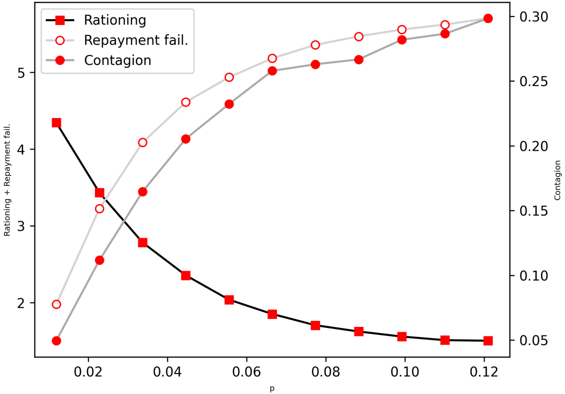
Monte Carlo simulations with $\mathcal{MC} = 50$ runs. $N = 50$ banks, $T = 1000$ periods, and p varying from 0 to 0.12 in increments of 0.01.

Initial conditions for each bank:

Assets	Liabilities
$C_0^i = 5 - 0.18 = 4.82$	$D_0^i = 9$
$L_0^i = 5$	$E_0^i = 1$
$R_0^i = 0.18$	

Bankruptcies decomposition

As p increases, the channel of failure shifts from rationing (few lenders available) to repayment failure (low rates, more clients served).

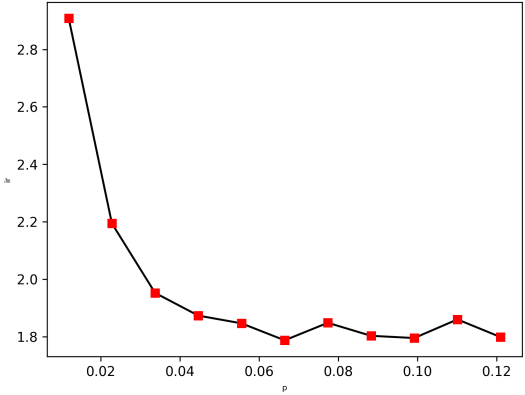


p	Rat.	Cont.	Fail.
0.012	4.35	0.05	1.98
0.023	3.43	0.11	3.23
0.034	2.78	0.16	4.09
0.045	2.36	0.21	4.61
0.056	2.04	0.23	4.94
0.066	1.85	0.26	5.18
0.077	1.71	0.26	5.36
0.088	1.63	0.27	5.47
0.099	1.56	0.28	5.56
0.110	1.51	0.29	5.62
0.121	1.51	0.30	5.71

Average number of failures during simulations.

Interest rate

As the probability of attachment p increases, the interest rate declines. This reflects a more competitive market: borrowers can choose among more lenders, reducing lenders' market power. Lower rates benefit borrowers but also compress lender margins and increase exposure to default.



p	r^i
0.012	2.9083
0.023	2.1940
0.034	1.9521
0.045	1.8731
0.056	1.8461
0.066	1.7871
0.077	1.8480
0.088	1.8027
0.099	1.7949
0.110	1.8598
0.121	1.7982

Average value for the interest rate of loans in the simulations.

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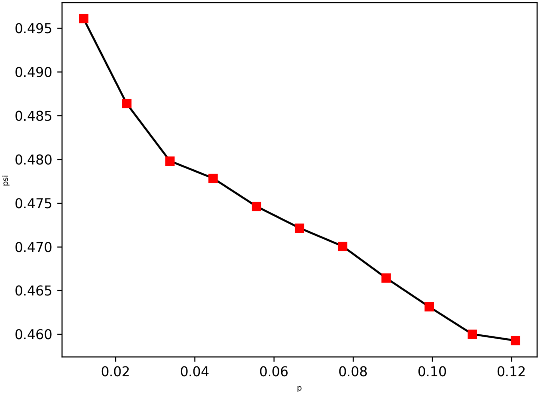
Market power ψ

Market power ψ decreases monotonically as connectivity p increases. In a denser network, the monopolistic markup is eroded. This is the mechanism through which connectivity drives interest rates downward.

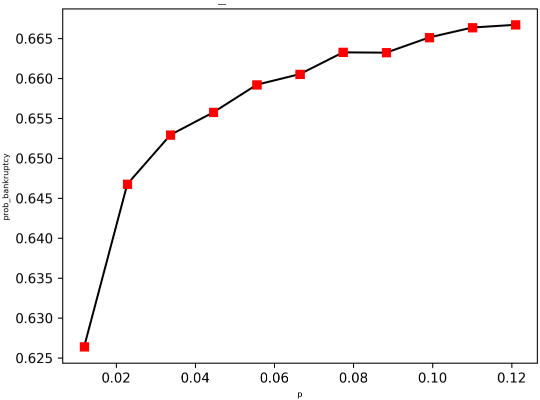


p	ψ
0.012	0.4961
0.023	0.4864
0.034	0.4798
0.045	0.4778
0.056	0.4746
0.066	0.4721
0.077	0.4701
0.088	0.4665
0.099	0.4631
0.110	0.4600
0.121	0.4593

Average value of market power of lenders in the simulations.

Probability of bankruptcy p_b

As connectivity increases, the probability of borrower bankruptcy rises. Lenders attempt to serve a larger number of clients and at the same time the system is being more fragile.

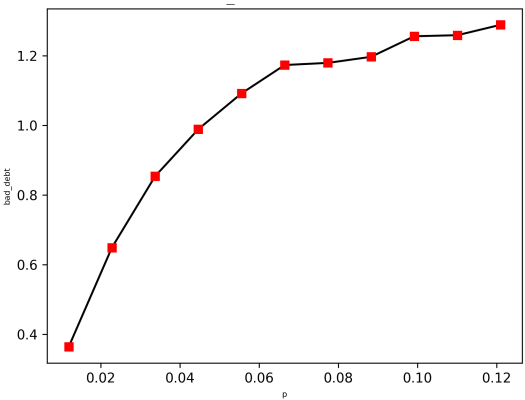


p	p_b
0.012	0.6264
0.023	0.6468
0.034	0.6529
0.045	0.6558
0.056	0.6592
0.066	0.6606
0.077	0.6633
0.088	0.6632
0.099	0.6652
0.110	0.6664
0.121	0.6667

Average value of p_b of borrowers in the simulations.

Bad debt

Bad debt grows with connectivity: more loans are extended as p increases, and a higher volume of lending translates into greater accumulated losses from default. This amplifies the systemic cost of low interest rate environments.



p	B
0.012	0.3636
0.023	0.6486
0.034	0.8536
0.045	0.9888
0.056	1.0917
0.066	1.1733
0.077	1.1794
0.088	1.1968
0.099	1.2559
0.110	1.2590
0.121	1.2888

Average sum of bad debt in the simulations.

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Findings (1/2)

As connectivity p increases:

1. Market power decreases with p, and the equilibrium interest rate falls also.
2. The probability of borrower bankruptcy rises, as lenders serve a larger number of clients.
3. Bad debt accumulates, reflecting greater systemic exposure to default.
4. Taken together, these results show that **an interest rate that remains too low for too long increases both the incidence of bankruptcies and the spread of contagion.**

Findings (2/2)

1. **Failure composition shifts**: rationing dominates at low p; repayment failure dominates at high p as interest rates decline.
2. The two opposing channels—lower rates from competition, and higher bankruptcy risk from greater connectivity—imply a **non-trivial trade-off** for regulators.
3. Excessively low interest rates, while beneficial for borrowers in the short run, can be **detrimental to the stability of the system** in the long run.

From here: reinforce learning for policy implications

- **Heterogeneity** is a root cause of systemic instability (failure of large banks).
- Interbank lending restricted among banks with **similar liquidity characteristics.**
- Reserve requirements dampen cascades but will **constrain growth.**
- **Monetary policy implications:** an interest rate environment that remains too low for too long can increase connectivity-driven fragility, suggesting that macroprudential policy must account for the endogenous response of network structure.